

PORVOO KALLBADAGRUND, Finland

WMO: 029870

Lat: 59.9853N

Lon: 25.5986E

Elev: 76

StdP: 14.66

Time Zone: 2.00 (EUE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	5.1	10.2	-1.5	5.1	8.7	3.0	6.4	12.5	41.9	14.7	38.5	24.8	14.3	50	0.892

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.4	73.6	67.7	71.3	66.0	69.0	63.9	68.9	72.3	66.6	70.1	64.4	67.6	13.1	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.5	101.3	71.6	65.1	93.0	69.1	62.9	86.1	66.9	33.2	72.5	31.3	70.1	29.6	67.5	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 39.4	35.6	32.3	DB	4.4	77.0	7.1	3.1	-0.7	79.3	-4.8	81.1	-8.8	82.8	-13.9	85.1	(4)
(5)			WB	3.3	69.0	6.9	2.2	-1.7	70.6	-5.7	71.9	-9.6	73.2	-14.6	74.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.1	27.5	25.9	30.8	38.1	48.5	55.9	63.4	63.0	56.3	46.3	39.0	33.6
	DBStd	14.17	7.99	8.80	5.87	4.26	6.12	4.58	4.82	4.13	4.32	5.47	5.99	6.79
	HDD50	3409	696	674	594	357	101	5	0	0	7	137	330	508
	HDD65	7683	1161	1094	1059	807	512	276	91	89	262	579	780	973
	CDD50	1274	0	0	0	0	56	180	416	403	196	23	0	0
	CDD65	71	0	0	0	0	1	2	41	27	0	0	0	0
	CDH74	56	0	0	0	0	2	1	41	12	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	17.5	19.6	17.4	17.5	15.8	15.4	15.6	14.3	15.0	16.6	19.6	21.1	22.2
Precipitation	PrecAvg	33.6	3.4	2.6	2.2	1.6	2.1	3.1	3.0	3.5	2.4	3.2	3.2	3.3
	PrecMax	47.9	7.1	5.6	4.4	4.0	3.9	6.7	6.4	8.0	4.3	5.5	5.2	5.6
	PrecMin	17.8	1.0	0.6	0.6	0.1	0.4	0.9	0.7	0.5	0.7	0.4	0.6	1.5
	PrecStd	6.8	1.5	1.3	1.1	0.9	1.0	1.4	1.4	1.7	1.0	1.2	1.3	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.1	39.3	46.4	53.1	69.3	70.4	77.8	75.7	66.4	57.6	50.4	46.8
		MCWB	40.2	35.6	39.3	45.2	54.8	60.6	68.1	69.3	62.3	55.0	48.7	45.1
	2%	DB	40.4	37.5	41.1	49.2	63.6	66.9	74.7	72.6	63.8	56.3	49.1	44.9
		MCWB	38.6	35.3	35.4	43.2	53.4	59.8	68.6	67.2	60.4	54.1	47.5	43.0
	5%	DB	39.0	36.4	38.7	45.9	59.7	64.1	72.6	70.4	62.6	55.1	47.6	43.2
		MCWB	37.5	34.5	34.5	41.3	52.7	57.9	67.4	65.0	59.3	52.9	45.9	41.2
	10%	DB	37.4	35.2	36.8	43.6	56.5	61.8	70.5	68.4	61.5	53.7	46.2	41.6
		MCWB	35.8	33.5	33.8	39.7	50.8	56.9	65.9	63.4	58.2	51.5	44.2	39.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.0	37.6	40.2	46.6	58.4	64.6	71.0	70.2	63.1	55.8	49.4	45.6
		MCDB	41.7	38.2	44.8	50.7	62.7	67.8	74.5	73.3	64.7	56.9	50.2	46.4
	2%	WB	39.1	36.1	37.6	44.0	55.6	61.8	69.8	67.4	61.5	54.6	47.8	43.4
		MCDB	39.9	36.9	39.4	46.7	59.8	65.3	73.3	71.2	63.3	56.0	48.8	44.6
	5%	WB	37.6	35.1	36.2	41.7	53.6	59.6	68.2	65.3	60.2	53.4	46.3	41.5
		MCDB	38.7	36.0	37.7	44.3	58.4	62.9	71.6	68.7	62.2	54.9	47.4	42.9
	10%	WB	36.1	34.1	35.0	40.1	51.5	57.8	66.2	63.8	58.9	51.8	44.7	39.8
		MCDB	37.3	35.0	36.3	42.6	56.2	61.0	69.6	66.8	61.2	53.5	45.9	41.3
Mean Daily Temperature Range	5% DB	MDBR	5.9	5.4	5.5	6.5	7.7	6.2	5.4	4.9	4.7	4.9	4.8	5.5
		MCDBR	5.4	4.4	6.9	10.3	12.0	9.5	7.2	6.4	5.1	4.9	4.8	5.3
		MCWBR	5.5	4.2	4.4	6.2	6.6	5.8	5.2	4.9	4.5	5.1	5.3	5.4
	5% WB	MCDWR	5.3	4.5	6.2	9.6	10.8	8.0	6.2	5.9	4.7	4.9	4.6	5.3
		MCWBR	5.6	4.6	4.6	6.6	7.6	5.9	5.3	5.2	4.7	5.2	5.3	5.6
Clear-Sky Solar Irradiance	taub	0.279	0.310	0.332	0.364	0.362	0.360	0.382	0.376	0.349	0.334	0.308	0.265	
	taud	2.149	2.182	2.223	2.307	2.410	2.478	2.439	2.452	2.512	2.399	2.220	2.104	
	Ebn at Noon	153	214	250	265	274	277	267	259	248	209	141	110	
	Edh at Noon	14	23	30	34	33	31	32	29	23	19	13	10	
All-Sky Solar Radiation	RadAvg	79	260	733	1223	1685	1752	1691	1313	798	345	95	47	
	RadStd	14	48	95	130	182	134	151	133	102	52	16	7	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (60 neighbors)		+0.77	N/A	N/A	N/A	N/A	N/A	-263	N/A	N/A	N/A	

Nomenclature: See separate page